PHONE HTET MYAT

Data Analyst & Engineer

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ABOUT ME

Pre-mid-level Data Engineer practitioner with a strong technical base in Python, SQL, and MLOps. Experience building and validating machine learning models, creating data pipelines, and deploying applications using Docker and FastAPI. A proactive problem-solver who enjoys bridging the gap between raw data and functional software. Seeking a challenging role to apply my technical skills, refine my production-level coding, and deliver value within a professional team.

EDUCATION

2022 - Present
University of Sunderland (partnership with GUSTO COLLEGE)

Bsc (Hons) Computer System Engineering

Currently attending in the final semester of final year, and concentration in Data Engineering, with advanced coursework in Machine Learning, predictive analytics, and statistical modeling.

2026- Present
University of the People

Associate of Science in Business Administration ((WSCUC Accredited)

Gaining a foundation in business strategy and operations to bridge the gap between technical data and corporate decision-making.

2022-2025
GUSTO COLLEGE

HND Diploma in Computing

Graduated with a strong technical foundation in software development and networking.

EXPERIENCE

2025-2025

Lead Developer/Product Owner of EASYEAT application

- Spearheaded the development of "Easy Eat," an application aimed at optimizing food service efficiency, from initial ideation to a functional prototype for a high-stakes innovation competition.
- Orchestrated a cross-functional team, managing the product roadmap, user experience design, and technical requirements to meet strict development deadlines.
- Presented technical architecture and business viability metrics to expert judges, demonstrating the ability to communicate complex data-driven solutions to non-technical stakeholders.

SKILLS & CERTIFICATIONS

Programming & Software Engineering

- Python
- Java
- R
- SQL
- Git
- pytest
- PEP 8 Standards
- OOP (Object-Oriented Programming)
- SOLID Principles
- Docker
- Linux / Bash
- FastAPI
- Excel
- Power BI

Data Science & Machine Learning

- Scikit-learn
- LinearRegression
- Logistic Regression
- Random Forest
- XGBoost
- K-Means Clustering
- PCA (Principal Component Analysis)
- PyTorch
- LangChain
- RAG (Retrieval Augmented Generation)
- Trend Forecasting
- KPI Tracking
- Business Intelligence

Data Engineering & Manipulation

- Pandas Polars
- NumPy
- ETL Pipelines
- Database Design
- Schema Design

Visualization & MLOps

- Matplotlib
- Seaborn
- Streamlit
- MLflow

Database and Storage

- My SQL
- Postgre SQL
- Oracle
- ChromDB

Certifications

- Applied Data Science Specialization (Dataversity)
- Data Science Lab Program (School of Matrix)
- Math Fundamentals for AI and Data Science (Simbolo)
- Data Science Program (Simbolo)
- Fundamental Python Programming (Dataversity)

Interpersonal & Business Skills

- Leadership
- Logical Thinking
- Creativity
- Teamwork
- Communication
- Problem Solving

TECHNICAL EXPERIENCE

Self Directed

1.Smart Expense Anomaly Detector
Technology: Python, Streamlit, Scikit-Learn, Pandas, Plotly, NumPy

Project Overview:
An intelligent financial monitoring tool designed to identify anomalous spending patterns in user transaction data using machine learning algorithms. The system provides automated anomaly detection, intuitive visualization of spending habits, and real-time transaction prediction.

Application:
- Data Engineering: Developed robust preprocessing pipelines using StandardScaler and LabelEncoder to normalize financial datasets and convert categorical spending data into model-ready formats.
- Model Development: Integrated multiple unsupervised learning algorithms, including Isolation Forest, One-Class SVM, and Local Outlier Factor, allowing users to configure contamination parameters to fine-tune sensitivity.
- Visualization: Architected an interactive Streamlit dashboard featuring real-time metrics, anomaly timeline mapping, and category-based spending distributions utilizing Plotly.
- Prediction Logic: Built a manual input interface that allows users to predict the status of individual transactions based on the trained model state.

Responsibility:
- Designed the end-to-end architecture from data ingestion to model inference.
- Implemented state management within Streamlit to ensure the persistence of model objects (st.session_state) across user interactions.
- Developed a comprehensive evaluation suite using precision_score, recall_score, and f1_score to validate model performance on test data.
- Focused on a modular code structure, enabling the addition of new algorithms and features without disrupting existing workflows.

Challenges and Solutions:
- Challenge: Implementing effective anomaly detection on unlabeled financial data without historical "fraud" tags.
- Solution: Engineered a ground-truth generation method by applying statistical thresholds (Mean + 2 Std Dev) to create synthetic labels for model training and validation.
- Challenge: Maintaining performance and model state within a reactive framework like Streamlit.
- Solution: Utilized st.session_state to cache the scaler and model objects, preventing redundant retraining cycles and ensuring the application remains responsive during user input.

Results:
- Successfully delivered a functional end-to-end machine learning application that empowers users to detect unusual financial activity with high precision.
- Achieved an intuitive UI design that simplifies complex machine learning concepts, making the data insights accessible for non-technical users.
- Optimized the application to handle real-time prediction inputs, providing immediate feedback on whether a transaction is "Normal" or "Anomalous."

Project Link : https://philip12-html-smart-expense-anomaly-detector.streamlit.app/

2.Credit Card Fraud Detection System

Technology: Python (Flask), Scikit-Learn, XGBoost, Logistic Regression, Random Forest (Tuned), Pandas, NumPy, HTML/CSS/JavaScript, SMOTE

Project Overview:
A high-performance anomaly detection system capable of processing financial transactions in real-time or via batch CSV uploads. The system utilizes an ensemble of machine learning algorithms to predict fraudulent activity, benchmarking multiple models to achieve the highest possible classification accuracy.

Application:

- ML Pipeline Development: Implemented a comprehensive ML pipeline using Scikit-Learn and XGBoost. Applied SMOTE (Synthetic Minority Over-sampling Technique) to address class imbalance and performed extensive Hyperparameter Tuning to optimize Random Forest performance.
- Model Benchmarking: Conducted a comparative analysis between Logistic Regression (baseline), standard Random Forest, and a Tuned Random Forest model to determine the most effective approach for detecting fraud.
- Backend Architecture: Built a RESTful API using Flask to handle asynchronous predictions, batch CSV processing, and real-time dashboard updates via JSON endpoints.
- Logic Layer: Engineered a "Smart Boost" heuristic system that combines model probability with rule-based features (e.g., transaction value, geographic distance, and time-of-day) to calculate a final "Risk Level."

Responsibility:

- Designed the full-stack architecture, integrating the ML backend with an interactive HTML/JS dashboard for real-time visualization of risk metrics.
- Developed fault-tolerant model loading routines to handle serialized assets (joblib) and resolved legacy versioning conflicts in XGBoost attributes.
- Established a state management system within the Flask application to track transaction logs and generate dynamic statistical reports.

Challenges and Solutions:

- Challenge: Financial datasets are inherently imbalanced (fraud is rare), leading to baseline models like Logistic Regression underperforming on the minority class.
- Solution: Integrated SMOTE to generate synthetic samples for the minority class and transitioned to ensemble methods (XGBoost and Tuned Random Forest) which significantly improved the recall rate compared to the baseline.
- Challenge: Models often struggle with non-linear patterns, such as high-value legitimate transactions being misclassified.
- Solution: Developed a post-processing logic layer that applies weighted "boosts" based on domain-specific heuristics, effectively reducing false-positive rates by 25%.
- Challenge: Deployment errors related to model versioning and missing file paths.
- Solution: Implemented robust defensive programming, including custom attribute cleanup for model deserialization to ensure cross-environment compatibility.

Results:

- Successfully delivered a production-ready system that outperformed the Logistic Regression baseline, with the Tuned Random Forest providing the optimal balance between precision and recall.
- Created a seamless user experience that allows non-technical users to process thousands of transactions via batch CSV uploads, receiving instant, human-readable risk insights.
- Optimized the application for sub-second inference times while maintaining high model interpretability through clear risk-level reporting.

Project Link : https://fraud-detection-a8c9.onrender.com